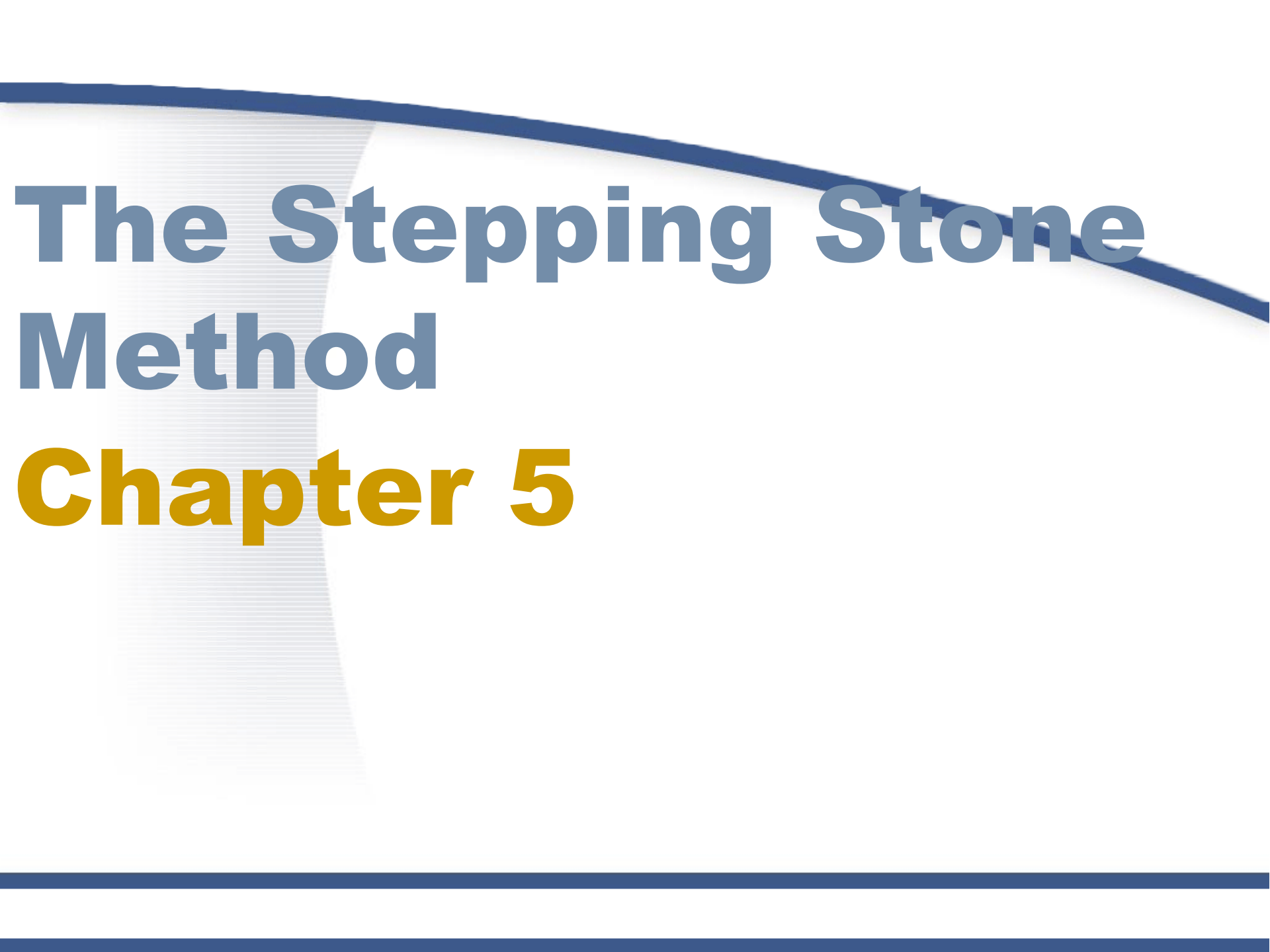




The Stepping Stone Method

Chapter 5

- ⑩ Once an initial basic feasible solution has been determined by any of the previous three methods, the next step is to solve the model for the optimal
- ⑩ Because the initial solution obtained by the minimum cell cost method had the lowest total cost of the three initial solutions, we will use it as the starting solution.
- ⑩ Check first : no. of occupied routes = no. of rows + no. of columns - 1
- ⑩ If less than this , the solution is called degenerate

The Stepping Stone Method

From \ To	A	B	C	Supply
1	6	8	10	150
2	7	11	11	175
3	4	5	12	275
Demand	200	100	300	600



10 Select an unused squares to be evaluated :

1A,2A,2B,3C

Beginning at this square trace a closed path back to the original squares via squares that are currently being used and moving with only horizontal, vertical moves

From \ To	A	B	C	Supply
1	+1 6	-1 8	10	150
2	7	11	11	175
3	-1 200	+1 75	5 12	275
Demand	200	100	300	600

Closed path : 1A $\rightarrow$ 1B $\rightarrow$ 3B $\rightarrow$ 3A
Improvement index : $+6 - 8 + 5 - 4 = -\$1$

INITIAL SOLUTION IS **NOT** OPTIMAL

From \ To	A	B	C	Supply
	6 - 8 + 10			
1		25	125	150
2	+ 7	11	- 11	175
3	- 4	+ 5	12	275
Demand	200	100	300	600

$2A \rightarrow 2C \rightarrow 1C \rightarrow 1B \rightarrow 3B \rightarrow 3A$
 $+\$7 - 11 + 10 - 8 + 5 - 4 = -\1

From \ To	A	B	C	Supply
1	6	8	10	150
2	7	11	11	175
3	4	5	12	275
Demand	200	100	300	600

$2B \rightarrow 2C \rightarrow 1C \rightarrow 1B$

$$+\$11 - 11 + 10 - 8 = +\$2$$

From \ To	A	B	C	Supply
	6 + 8 - 10			
1		25	125	150
2	7	11	11	175
			175	
3	4 - 5 + 12	75		275
	200			
Demand	200	100	300	600

$$3C \rightarrow 1C \rightarrow 1B \rightarrow 3B$$

$$+\$12 - 10 + 8 - 5 = +\$5$$

The Stepping-Stone Path for Cell 1A

From \ To	A	B	C	Supply
1	+ 6	- 8	10	150
2	7	11	11	175
3	- 4	+ 5	12	275
Demand	200	100	300	600

The Second Iteration of the Stepping-Stone Method

From \ To				Supply
	A	B	C	
1		6	10	150
	25		125	
2		7	11	175
			175	
3		4	12	275
	175	100		
Demand	200	100	300	600

From \ To	A	B	C	Supply
	- 6	8	+ 10	
1	25		125	150
	+ 7	11	- 11	
2			175	175
	4	5	12	
3	175	100		275
Demand	200	100	300	600

$2A \rightarrow 2C \rightarrow 1C \rightarrow 1A$

$$+\$7 - 11 + 10 - 6 = \$0$$

From \ To	A	B	C	Supply
1	- 25	+ 8	10	150
2	7	11	11	175
3	+ 175	- 100	12	275
Demand	200	100	300	600

$1B \rightarrow 3B \rightarrow 3A \rightarrow 1A$

$$+\$8 - 5 + 4 - 6 = +\$1$$

From \ To	A	B	C	Supply
1	- 25	8	+ 10	150
2	7	+ 11	- 11	175
3	+ 4	- 5	12	275
Demand	200	100	300	600

$2B \rightarrow 3B \rightarrow 3A \rightarrow 1A \rightarrow 1C \rightarrow 2C$
 $+\$11 - 5 + 4 - 6 + 10 - 11 = +\3

From \ To	A	B	C	Supply
1	+ 6 25	8	- 10 125	150
2	7	11	11 175	175
3	- 4 175	5 100	+ 12	275
Demand	200	100	300	600

$3C \rightarrow 3A \rightarrow 1A \rightarrow 1C$

$$+\$12 - 4 + 6 - 10 = +\$4$$

$$x_{1A} = 25 \text{ tons}$$

$$x_{2C} = 175 \text{ tons}$$

$$x_{3A} = 175 \text{ tons}$$

$$x_{1C} = 125 \text{ tons}$$

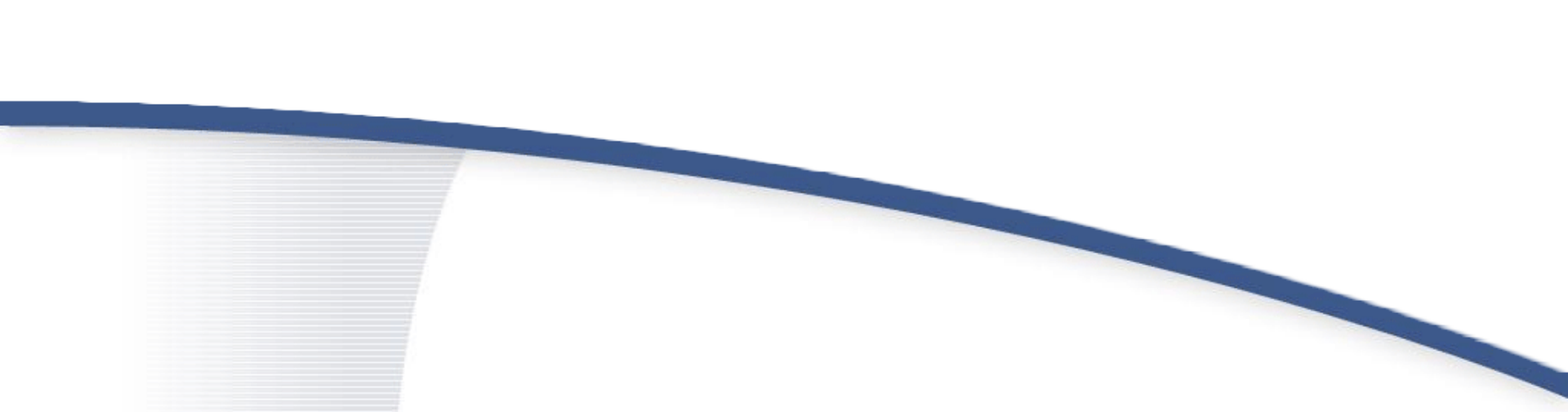
$$x_{3B} = 100 \text{ tons}$$

$$\begin{aligned} Z &= \$6(25) + 8(0) + 10(125) + 7(0) + 11(0) + 11(175) + 4(175) + 5(100) + 12(0) \\ &= \$4,525 \end{aligned}$$

The Alternative Optimal Solution

2A with zero improvement index after we found optimal solution

From \ To	A	B	C	Supply
1	6 150	8 150	10 150	150
2	7 25	11 150	11 150	175
3	4 175	5 100	12 275	275
Demand	200	100	300	600

- 
1. Determine the stepping-stone paths and cost changes for each empty cell in the tableau.
 2. Allocate as much as possible to the empty cell with the greatest net decrease in cost.
 3. Repeat steps 1 and 2 until all empty cells have positive cost changes that indicate an optimal solution.

m rows + n columns $- 1 =$ the number of cells with allocations

The Minimum Cell Cost Initial Solution

From \ To	A	B	C	Supply
1	6 100	8 50	10 150	
2	7 250	11 250	11 250	
3	4 200	5 200	12 200	
Demand	200	100	300	600

$m + n - 1 =$ the number of cells with allocations

$$3 + 3 - 1 = 5 \text{ cells}$$

The Initial Solution

From \ To				Supply
	A	B	C	
1	6	8	10	150
	0	100	50	
2	7	11	11	250
			250	
3	4	5	12	200
	200			
Demand	200	100	300	600

$$x_{2A}: \begin{array}{cccc} 2A & 2C & 1C & 1A \\ 7 & -11 & +10 & -6 = 0 \end{array}$$

$$x_{2B}: \begin{array}{cccc} 2B & 2C & 1C & 1B \\ 11 & -11 & +10 & -8 = +2 \end{array}$$

$$x_{3B}: \begin{array}{cccc} 3B & 1B & 1A & 3A \\ 5 & -8 & +6 & -4 = -1 \end{array}$$

$$x_{3C}: \begin{array}{cccc} 3C & 1C & 1A & 3A \\ 12 & -10 & +6 & -4 = +4 \end{array}$$

The Second Stepping-Stone Iteration

From \ To	A		B		C		Supply
1		6		8		10	150
	100				50		
2		7		11		11	250
					250		
3		4		5		12	200
	100		100				
Demand	200		100		300		600

Prohibited Routes

- 10 Sometimes one or more of the routes in the transportation model are prohibited. That is, units cannot be transported from a particular source to a particular destination. When this situation occurs, we must make sure that no units in the optimal solution are allocated to the cell representing this route. In our study of the simplex tableau, we learned that assigning a large relative cost or a coefficient of m to a variable would keep it out of the final solution. This same principle can be used in a transportation model for a prohibited route. A value of m is assigned as the transportation cost for a cell that represents a prohibited route. Thus, when the prohibited cell is evaluated, it will always contain a large positive cost change of m , which will keep it from being selected as an entering variable.
- 10 A prohibited route is assigned a large cost such as m so that it will never receive an allocation.

Example

Consider the following transportation problem.

From	To			Supply
	1	2	3	
A	\$ 6	9	M	130
B	12	3	5	70
C	4	8	11	100
Demand	80	110	60	

- Find the initial solution by using VAM and then solve it using the stepping-stone method.
- Formulate this problem as a general linear programming model.

Example # 2

10 Solve the following linear programming problem.

minimize $Z = 3x_{11} + 12x_{12} + 8x_{13} + 10x_{21} + 5x_{22} + 6x_{23} + 6x_{31} + 7x_{32} + 10x_{33}$
subject to

$$x_{11} + x_{12} + x_{13} = 90$$

$$x_{21} + x_{22} + x_{23} = 30$$

$$x_{31} + x_{32} + x_{33} = 100$$

$$x_{11} + x_{21} + x_{31} \leq 70$$

$$x_{12} + x_{22} + x_{32} \leq 110$$

$$x_{13} + x_{23} + x_{33} \leq 80$$

$$x_{ij} \geq 0$$